

TEZPUR UNIVERSITY
Autumn Semester Mid Examination, 2025

Course Code: CSMT 530

Course Title: Mathematical Foundation for Data Science

Full marks: 30

Date: 10/10/2025

Time: 1.5 hours

All questions are compulsory

Q1. Write true or false:

$1 \times 5 = 5$

- (i) For $Ax = b$, the vectors b that are not in the column space of A form a subspace. $\narrow$
- (ii) The column space of $2A$ equal to the column space of A .
- (iii) An invertible matrix has no free variables. $\narrow$
- (iv) The column space of a matrix A has the same dimension as its row space. $\narrow$
- (v) The determinant of $AB - BA$ is zero.

Q2.

$2 + 3 + 4 + 2 = 11$

- (i) Choose a coefficient b that makes this system singular. Then choose a right hand side g that makes it solvable.

$$2x + by = 16$$

$$4x + 8y = g$$

- (ii) Solve the linear system:

$$x + 4y + 2z = -2$$

$$-2x - 8y + 3z = 32$$

$$y + z = 1$$

- (iii) Derive the reduced row echelon form of $A = \begin{bmatrix} 2 & 4 & 6 & 4 \\ 2 & 5 & 7 & 6 \\ 2 & 3 & 5 & 2 \end{bmatrix}$. Solve $Ax = b$, for

$$b = \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}.$$

- (iv) Describe the subspace of R^3 spanned by the three vectors $(0,1,1)$, $(1,1,0)$ and $(0,0,0)$.

Q3.

$3 + 3 = 6$

- (i) Project the vector $b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ onto the line through $a = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

- (ii) Apply Gram-Smith process to make the matrix $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ orthonormal.

Q4.

- (ii) Compute the determinant of $A = \begin{bmatrix} 4 & 4 & 4 & 4 \\ 1 & 2 & 0 & 1 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \end{bmatrix}$.

- $$\begin{array}{rcl} x + 4y + z & = & 1 \\ x + y + z & = & 0 \\ 2x + 3z & = & 0 \end{array}$$

$$\begin{bmatrix} 1 & 4 & 2 \\ 1 & 1 & 1 \\ 2 & 0 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 2 \\ 0 & -3 & -1 \\ 0 & -4 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 2 \\ 0 & -3 & -1 \\ 0 & 0 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 7 \\ 0 & -3 & -1 \\ 0 & 0 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 1/3 \\ 0 & 0 & 5 \end{bmatrix}$$

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$$\rightarrow L - n + 2$$

7-9

Question No	Q1	Q2	Q3	Q4
Course Outcome	CO1	CO3	CO1	CO1
Bloom Taxonomy	L2	L3	L2	L2

COs	Statements	Blooms Level
CO1	Explain basic components of linear algebra used in data science and computer science.	L2
CO2	Explain basic components of probability and statistics used in data science and computer science.	L2
CO3	Develop pattern recognition techniques for solving real-world problems.	L3

Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3		2				1			1			1
CO2	3	3	3		2				1			1			1
CO3	3	3	3	3	2				1			2	1		1

TEZPUR UNIVERSITY
Autumn Semester Test 1 Examination, 2025
CSMT 500: Mathematical Foundation for Computer Science
CSMT 530: Mathematical Foundation for Data Science

Full Marks: 10

Date: 13/09/2025

Time: 40 Minutes

Attempt all questions.

Q1. Under what condition on y_1, y_2, y_3 do the points $(0, y_1), (1, y_2), (2, y_3)$ lie on a straight line? 1

Q2. Choose a right-hand side which gives no solution and another right-hand side which gives infinite many solutions for the following. 2

$$3x + 2y = 10$$

$$6x + 4y = ?$$

Q3. Use elimination to solve 2

$$u + v + w = 16$$

$$u + 2v + 2w = 11$$

$$2u + 3v - 4w = 3$$

Q4. True or False? 1

(i) If rows 1 and 5 of A are the same, so are rows 1 and 5 of AB.

Q5. Find the inverse of $\begin{bmatrix} 1 & 4 \\ 3 & 9 \end{bmatrix}$ using Gauss-Jordan method. 2

Q6. Find the column, row, null and left null spaces of $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$. 2

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TEZPUR UNIVERSITY

Autumn End Semester Examination, 2025

Course Code: CSMT 530

Course Title: Mathematical Foundation for Data Science

Full Marks: 50

Date: 12/12/2025

Time: 2 hours

All questions are compulsory

Q1.

1+3+3+3=10

- i. Find the determinant of

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 1 \\ 1 & 4 & 1 & 1 \end{bmatrix}$$

- ii. Find the Eigen value and Eigen vector of

$$\begin{bmatrix} 1 & 4 & 5 \\ 0 & \frac{3}{4} & 6 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$$

- iii. If $A = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$, find A^{1000} .

- iv. If the Eigen values of A are 1, 1, 2, which of the following are certain to be true? Give reason.

- A is invertible.
- A is diagonalizable.
- A is not diagonalizable.

Q2.

4+3+3=10

- Find the 100th Fibonacci number.
- Compute the length and the inner product of $x = \begin{bmatrix} 2 - 4i \\ 4i \end{bmatrix}$ and $y = \begin{bmatrix} 2 + 4i \\ 4i \end{bmatrix}$.
- Decompose $\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ into sum of rank-one matrix.

Q3.

3+3+4=10

- i. For what range of number a , the matrix A is positive definite?

$$A = \begin{bmatrix} a & 2 & 2 \\ 2 & a & 2 \\ 2 & 2 & a \end{bmatrix}$$

- Find the SVD of singular matrix $\begin{bmatrix} 1 & 4 \\ 2 & 8 \end{bmatrix}$.
- Find the SVD of rank-two matrix $\begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$.

Q4.

3+4+3=10

- i. Two dice are rolled, if it is known that at least one of the dice always shows 4, find the probability that the numbers appeared on the dice have a sum 8.
- ii. A bag contains 3 red and 7 black balls. Two balls are drawn at random without replacement. If the second ball is red, what is the probability that the first ball is also red?
- iii. A fair coin is tossed 3 times. Let the random variable X denote the number of heads in 3 tosses of the coin. Find the sample space, the space of the random variable, and the probability density function of X.

Q5.

4+3+3=10

- i. Let X be a continuous random variable with following probability density function: $f(x) = \begin{cases} ce^{-x} & x \geq 0 \\ 0 & \text{otherwise,} \end{cases}$
 - (a) Find c
 - (b) Find the cumulative distribution function of X,
 - (c) Find $P(1 < X < 3)$
- ii. The probability distribution of a discrete random variable X is given by $f(x) = \begin{cases} k(2-x)^2 & x = -2, -1, 0, 1, 2 \\ 0 & \text{otherwise,} \end{cases}$
 - (a) Determine the value of K.
 - (b) Find the value of $E(X)$ and $E(X^2)$.
- iii. If X is a random variable with mean μ_X and variance σ_X^2 , then show that $\sigma_X^2 = E(X^2) - (\mu_X)^2$.

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